

Tarvin Kaur Arora

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EDUCATION

Bachelor of Science in Statistics (Expected Graduation: May 2027)

University of British Columbia

WORK EXPERIENCE

Undergraduate Teaching Assistant | University of British Columbia | September 2026 – Present

- Lead two in-person weekly lab sessions for an introductory computational-thinking course, guiding students through hands-on exercises in data representation, algorithms, and foundational programming to build problem-solving skills.
- Supported course delivery alongside the teaching team by grading student work with constructive feedback and contributing to lab and course-project materials, reinforcing core concepts and improving learning outcomes.

Data Science Intern | Marsh McLennan (Mercer Consulting Private Limited) | June 2025 – August 2025

- Built an InfoSec query chat assistant using OCR-based field extraction from client documents, enabling automated Q&A retrieval and reducing client-legal query turnaround from 14 days to near-instant retrieval across 1,210 extracted fields.
- Developed a Python workflow to generate structured knowledge documents for a RAG-enabled chatbot, leveraging document chunking, vector embeddings, and retrieval guardrails to constrain responses to approved client documents.
- Conducted end-to-end UAT testing for chatbot features, validating retrieval quality, edge-case handling, user flows, and response consistency while documenting defects, user guides, FAQs, and release notes to support internal deployment.
- Developed workflow-based Excel macro automation for project tracking, synchronizing milestones and status updates across teams while reducing manual update effort by 60–70%.
- Developed a Power BI compensation benchmarking dashboard with 12+ KPIs using Power Query transformations, DAX measures, and dynamic visuals for remuneration analysis.

STEM and Language Tutor | Varsity Tutors | January 2024 - Present

- Tutored 50+ students in statistics, mathematics, and computer science, strengthening foundations in probability, data analysis, programming logic, and quantitative problem-solving.
- Supported college-level analytics and statistics coursework involving EDA, multivariate linear regression, hypothesis testing, residual analysis, model evaluation, data visualization, and debugging internship-related projects.
- Delivered real-time technical support across STEM subjects while maintaining a 5-star rating record through clear communication and adaptive teaching.

TECHNICAL PROJECTS

Factorial Experiment Analysis of Neural Network Performance

Technologies: Python, NumPy, Pandas, Torch, scikit-learn, ANOVA

- Designed and executed a factorial experiment on simulated data to evaluate 20+ combinations of preprocessing strategies, model architecture, and capacity regimes across 5000 simulated data points in a combination of numerical and categorical features.
- Built, trained, and compared MLP, CNN, and LSTM models in PyTorch, applying scikit-learn for preprocessing, train-test evaluation, and performance metrics including accuracy and F1 score across Latin Square, fractional, and full factorial designs.
- Conducted ANOVA-based statistical inference to interpret main effects, interaction effects, and model performance differences, validating assumptions with residual diagnostics and Q-Q plots.

Seismic Signal Characterization Using Unsupervised Learning

Technologies: Python, MiniSom, K-means, scikit-learn, PCA, FFT, cosine similarity

- Preprocessed multi-component seismic waveform windows across north-south, east-west, and vertical channels and extracted Fast Fourier Transform-based spectral features for frequency-domain analysis.
- Applied Self-Organizing Maps and K-Means clustering to group high-dimensional seismic spectra by frequency-pattern similarity across earthquakes and monitoring stations.
- Evaluated cluster similarity across 3 different earthquakes using cosine similarity (0.85-0.97), PCA projections, SOM hit maps, and cluster-occupancy distributions, identifying stable station-level frequency responses and event-specific variation.

Streaming Platform Analytics Database Design (SQL vs. NoSQL)

Technologies: SQL, MongoDB, Python, PyMongo, Pandas, Matplotlib

- Designed ER diagrams, relational SQL schemas and non-relational MongoDB schemas for Netflix and YouTube streaming datasets, modeling platform, content, engagement, pricing, and time-period entities.
- Developed SQL queries and MongoDB aggregation pipelines to transform, integrate, and analyze pre-, during-, and post-pandemic streaming trends across platform performance and user engagement.
- Conducted Python-based hypothesis testing to evaluate pandemic-period engagement shifts, and compared SQL vs. NoSQL trade-offs in schema flexibility, query complexity, data integrity, and analytical reporting.

TECHNICAL SKILLS

Programming Languages: Python, R, Java

Databases: SQL, MongoDB

Data Science & Analysis: R (tidyverse, ggplot2, dplyr, glmnet), Python (pandas, NumPy, matplotlib, seaborn, scikit-learn), Excel

Software & Tools: Power BI, Tableau, Git/Github, VS Code

OCR & Automation Frameworks: PyTesseract, pdf2image, FuzzyWuzzy